Protocols – Dialysis of IMAC Eluates (DHAK2 and PPK2)

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

This protocol dialyses the IMAC eluates of DHAK2 and PPK2 (Protocol_ Immobilised Metal Affinity Chromatography) to remove imidazole and excess salts and to exchange the enzymes into a low-concentration sodium phosphate buffer. The eluate is placed in a semi-permeable dialysis bag that retains the protein. Small molecules such as imidazole and NaCl diffuse out into a large volume of buffer until both sides reach equilibrium. After an overnight dialysis, the enzymes are in low-concentration sodium phosphate buffer (5 mM), ready either for the activity assays or for freeze-drying.

Materials list

Reagents

- IMAC eluate of DHAK2 or PPK2.
- Sodium phosphate, dibasic: $Na_2HPO_4 \cdot 2H_2O$ (177.99 g/mol).
- HCl and NaOH (1 M), for pH adjustment.
- Ultrapure water (Milli-Q).

Consumables

- Dialysis tubing (roll), with a molecular weight cut-off (MWCO) below the molecular weight of the protein.
- Dialysis clips (optional).
- Pipette or small funnel.
- Collection tubes or vials for the dialysed sample.

Equipment

- 3 L flask or beaker.
- Magnetic stirrer and stir bar.

Steps

A. Buffer preparation:

The dialysis buffer only needs to keep the pH stable, so its exact concentration is not critical. We used **5 mM sodium phosphate, pH 8.0** by default, so that the sample could be freeze-dried afterwards if needed (freeze-drying concentrates every salt left in the sample).

A slightly higher concentration (e.g. 15 mM, from a buffer already prepared) is also suitable if the sample goes directly to the activity assays. To avoid weighing and handling very large volumes, prepare 300 mL of a 10× stock (50 mM sodium phosphate, pH 8.0) and dilute it 1/10 with Milli-Q water to 3 L.

Note: Adjust the pH of the 10× stock before making it up to its final volume.

B. Dialysis bag preparation:

1. Cut a piece of dialysis tubing from the roll (length depends on the volume of sample). Add a few extra centimetres at each end for the clips, or more if you will close the bag with knots, and leave enough free space inside for the sample to swell slightly during dialysis.
2. Hydrate the tubing by soaking it in Milli-Q water in a flask for 15–20 min.
3. Open the tubing by rubbing it gently between your fingers, and rinse the inside with Milli-Q water so that it opens completely.
4. Close one end with a dialysis clip or a double knot.
5. Fill the bag with Milli-Q water and squeeze it gently to check that there are no leaks. Then empty it completely.
6. Load the sample (the IMAC eluate) into the bag with a pipette or a small funnel. Pre-wet the pipette tip or funnel with Milli-Q water, so that the sample flows smoothly and the tip does not stick to and tear the membrane.
7. Gently press out as much air as possible, leaving a little free space above the sample, and close the other end with a clip or a knot.

Handle the membrane gently at all times. A tear or a poorly closed end will lose the whole sample into the buffer. ⚠️

C. Dialysis:

1. Immerse the closed bag in the 3 L of pre-cooled dialysis buffer. Check that the bag floats freely and cannot be hit by the stir bar.
2. Stir gently **overnight** at 4 °C (cold room). Stirring keeps the buffer around the bag homogeneous, so diffusion continues at a steady rate.
3. The next day, remove the bag from the buffer.
4. Hold the bag upright and open one end, by removing the clip or cutting just below the knot.
5. Transfer the sample with a pipette or funnel into a suitable container (vial, Eppendorf tube or test tube, depending on the volume). Note the final volume, which may be slightly larger than the starting volume.
6. Keep the sample on ice or at 4 °C and use it directly for the activity assays, or proceed to freeze-drying.

Troubleshooting

- **Sample lost from the bag:** Check the tubing for leaks with Milli-Q water before loading the sample (step 5), make sure the clips or knots are tight, and keep the bag away from the stir bar.

- **Sample volume increased after dialysis:** Water enters the bag when the sample has a higher solute concentration than the buffer, which is common with eluates containing 1 M imidazole. Leave some free space in the bag when closing it. If the sample becomes too dilute, concentrate it (e.g. with centrifugal concentrators) or elute the IMAC in a smaller volume.

- **Residual imidazole still affecting the activity assay:** A single overnight dialysis against 3 L may not remove all of it, especially from eluates with 1 M imidazole. Change the buffer at least once (e.g. 2–3 h in the first 1.5 L, then overnight in a fresh 1.5 L), which removes much more imidazole than a single bath of the same total volume.